



# LLMs Explained for Beginners From Theory to Math



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# What is an LLM

## Definition:

A Large Language Model is a type of AI trained on massive amounts of text to understand and generate human-like language.

**Example:** ChatGPT, Claude, Gemini.

## Key Idea:

- Predicts the next word in a sequence.
- Learns grammar, facts, reasoning from data.

## Formula:

$$P(w_t \mid w_1, w_2, \dots, w_{t-1})$$

Meaning: Probability of the next word given all previous ones.



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## 02 How Do They Work? (Big Picture)

- **Tokenization** → Split text into units (words, subwords, characters).
- **Embedding** → Convert tokens into vectors (numbers).
- **Neural Network** → Processes these vectors through layers to find patterns.
- **Prediction** → Output probabilities for the next token.
- **Decoding** → Pick the next word and repeat.



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## The Model Architecture

**Most LLMs use Transformers.**

**Key components:**

- Encoder/Decoder blocks
- Self-Attention: learns which words relate to others.
- Feedforward layers: process features after attention.

**Core Math: Self-Attention**

- For each token:

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V$$

- Q = Queries
- K = Keys
- V = Values
- dk\_t = vector dimension for scaling.



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## 05 Why “Large”

- Parameters: numbers the model learns (like neuron weights).
- GPT-3 → 175 billion parameters
- More parameters = more capacity, but more compute.
- Size helps capture complex patterns in language.



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## The Math Recap

- Probability → Predicts next token.
- Vectors → Represent words in numerical space.
- Matrix Multiplication → Core of neural networks.
- Softmax → Converts raw scores to probabilities.
- Gradient Descent → Adjusts parameters to improve predictions.



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# 07 Applications

- Chatbots
- Code generation
- Content summarization
- Language translation
- Data analysis



# 08

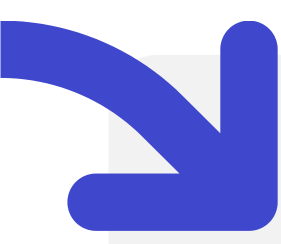
## Limitations & Risks

- Hallucinations (makes up facts)
- Bias from training data
- Compute costs
- Not truly “understanding” language



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LLMs are powerful tools trained to  
predict words

the magic comes from a mix of  
probability, linear algebra, and  
enormous amounts of data.

Follow for more AI breakdowns 



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